

Project Report: Train Ticket Booking System

1. Project Objective

The objective of this project is to build a console-based Train Ticket Booking interface using Python. The system simulates the basic workflow of a railway reservation portal, allowing users to navigate through booking, status checking, and cancellation options. It serves to demonstrate fundamental programming concepts including infinite loops, conditional branching, and basic user input validation.

2. Technology Stack

- **Language:** Python 3.x
- **Environment:** Command-line Interface (Terminal)
- **Dependencies:** Core Python only (No external libraries required)

3. System Modules

The project is divided into four main functional blocks accessed via a main menu loop.

- **Module 1: booking.py**
Collects passenger demographics (Name, Age, Gender) and journey details (Departure, Destination, Train Number). It prompts the user to select from five seating classes (First Class AC, AC Two Tier, AC Three Tier, Sleeper, General). It includes a validation check to ensure the train number entered is exactly 5 digits long before confirming the booking.
- **Module 2: reservation.py**
Evaluates a ticket's confirmation probability. The user inputs a 10-digit PNR and their current waitlist number. Based on the waitlist integer, the system categorizes the confirmation chances into four tiers (Above 90%, Above 50%, Above 20%, or Below 20%).
- **Module 3: cancellation.py**
Simulates the cancellation process by requesting the 5-digit train number, a 4-digit booking ID, route details, and the travel class. It validates the length of the IDs and outputs a successful cancellation and refund generation message.

4. Implementation Details

- **Control Flow:** The program utilizes a while 2>1: loop to keep the terminal running continuously until the user selects the "Exit" option.

- **Validation:** Basic range checking is implemented for numerical inputs. For example, a 5digit train number is validated using if $x < 10000$ or $x > 100000$..
- **Data Structure:** A Python dictionary ($s = \{\text{'name': p, 'age': q, 'gender': r}\}$) is used briefly to group the passenger's personal details during the booking sequence.

5. Output and Execution Screenshots

Below is the execution flow of the system.

Main Menu & Booking Process:

```
Welcome to online train ticket booking system!
1. Online Ticket Booking
2. Check Current Reservation
3. Ticket Cancellation
4. Exit
What you wish to do today: 1
Enter your name: Parth
Enter your age: 17
Enter your gender: M
Enter your departure station: BPL
Enter your destination station: NDLS
Enter 5 digit train number: 67676
1. First Class AC
2. AC Two tier
3. AC Three tier
4. Sleeper Class
5. General Class
Enter preferred class: 2
{'name': 'Parth', 'age': 17, 'gender': 'M'}
Your ticket booking with train number 67676 from BPL to NDLS in AC Two Tier has been confirmed!
```

PNR Status Check:

```
Welcome to online train ticket booking system!
1. Online Ticket Booking
2. Check Current Reservation
3. Ticket Cancellation
4. Exit
What you wish to do today: 2
Enter your 10 digit PNR number: 6767676767
Enter your last waiting list number:67
Your ticket with PNR 6767676767 has confirmation rate above 50%
Thank You!
```

Ticket Cancellation:

```
Welcome to online train ticket booking system!
1. Online Ticket Booking
2. Check Current Reservation
3. Ticket Cancellation
4. Exit
What you wish to do today: 3
Your train number: 12345
Enter your 4 digit booking ID: 6767
Enter your departure station: BPL
Enter your arrival station: NDLS
1. First Class AC
2. AC Two tier
3. AC Three tier
4. Sleeper Class
5. General Class
Enter your seat class: 2
Your ticket with train number 12345 and booking ID 6767 from BPL to NDLS in AC Two tier has been canceled successfully!!
Refund will be generated shortly
Thank You!
```

6. Limitations and Future Scope

Limitations

- **Stateless Execution:** The system does not use a database or file handling. Variables are stored temporarily in memory and reset once the loop restarts or the program closes.
- **Type Exceptions:** The program expects numerical inputs for age, train number, and PNR. Entering alphabetic characters in these fields will cause a runtime ValueError.

Future Scope

- Integrate a SQLite database or CSV file handling to store and retrieve actual user bookings.
- Implement try-except blocks to handle string-to-integer conversion errors gracefully without crashing the program.
- Add a module to display actual train routes, timings, and dynamic seat availability.